

Parabolic Inverse Source Reconstruction: From Raw Error to Visible Geometry

Pingsen Chow (supervised by Prof. Zhiwen Zhang)
Faculty of Science, The University of Hong Kong

Summer Research Fellowship
(SRF) 2025 for Science Students

Poster No.: A11
Name: CHOW, Ping Sen
University No.: 3036450847
Student's Major: Mathematics

00 The inverse problem and POD reconstruction

The inverse problem

$$\begin{cases} \partial_t u + Lu = f^\dagger, & (x, t) \in \Omega \times (0, T), \\ u = 0, & (x, t) \in \partial\Omega \times (0, T), \\ u(\cdot, 0) = 0. \end{cases}$$
$$f \xrightarrow{S_T} u_f(\cdot, T), \quad M^\delta = S_T f^\dagger + \xi.$$
$$f_\lambda^\delta = \arg \min_f \left\{ \|S_T f - M^\delta\|_{L^2}^2 + \lambda \|f\|_{L^2}^2 \right\}.$$

Diffusion suppresses fine spatial information, making source recovery from final-time data ill posed.

01 Three levels of reconstruction error

Declare a physical visibility scale $\ell > 0$ and set $H_\ell = e^{-\ell^2 L}$.

RAW SOURCE	$E_{\text{raw}} = \frac{\ f_{\text{rec}} - f^\dagger\ _{L^2}}{\ f^\dagger\ _{L^2}}$
VISIBLE FIELD	$E_{\text{vis},2}(\ell) = \frac{\ H_\ell(f_{\text{rec}} - f^\dagger)\ _{L^2}}{\ H_\ell f^\dagger\ _{L^2}}$
VISIBLE GEOMETRY	$E_{\text{geom}} = d_H(\Gamma_\tau(H_\ell f_{\text{rec}}), \Gamma_\tau(H_\ell f^\dagger))$

Numerics report relative L^2 visible error for average mismatch at scale ℓ . Geometry stability instead requires pointwise L^∞ control of the same heat-smoothed discrepancy.

02 The visibility scale

Let $L\phi_j = \mu_j\phi_j$. Tikhonov regularization has spectral gain

$$s_T(\mu) = \frac{1 - e^{-T\mu}}{\mu}, \quad g_\lambda(\mu) = \frac{s_T(\mu)^2}{s_T(\mu)^2 + \lambda}.$$

With $c_T = 1 - e^{-T\mu_1}$, any half-gain crossing on $[\mu_1, \infty)$ satisfies

$$c_T \lambda^{-1/2} \leq \mu_{1/2} \leq \lambda^{-1/2}, \quad [\ell_{\lambda,T} = c_T^{-1/2} \lambda^{1/4}].$$

The scale $\ell_{\lambda,T}$ is conservative rather than an exact cutoff; structures substantially below it are strongly suppressed.

03 Raw versus visible recovery

The spectral projector P_r retains the first r eigenmodes of L ; it is distinct from the learned POD projector. For a two-dimensional Lipschitz indicator source,

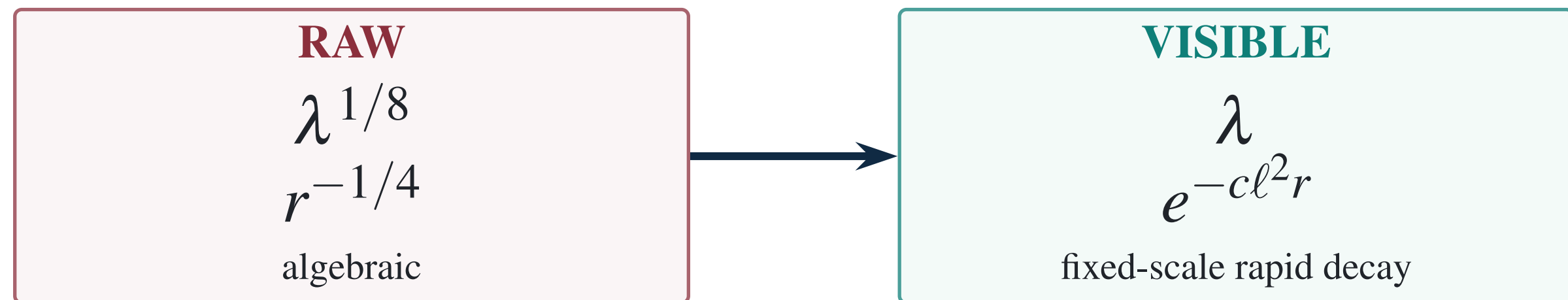
$$\|f_{\lambda,r}^\delta - f^\dagger\|_{L^2} \lesssim \frac{\delta}{\sqrt{\lambda}} + \lambda^{1/8-\varepsilon} + r^{-1/4+2\varepsilon}.$$

With exact data, fixed $0 < \ell \leq \sqrt{T}$, and $\ell_{\lambda,T} \leq \ell$, the actual pointwise discrepancy satisfies

$$e_{\text{vis},\infty} := \|H_\ell(f_{\lambda,r}^0 - f^\dagger)\|_{L^\infty} \leq \varepsilon_{\text{vis}},$$
$$\varepsilon_{\text{vis}} \lesssim C_{\ell,T,f^\dagger} \left(\lambda + e^{-c\ell^2 r} \right).$$

Here ε_{vis} is the theoretical visible-error budget.

KEY RATE SEPARATION

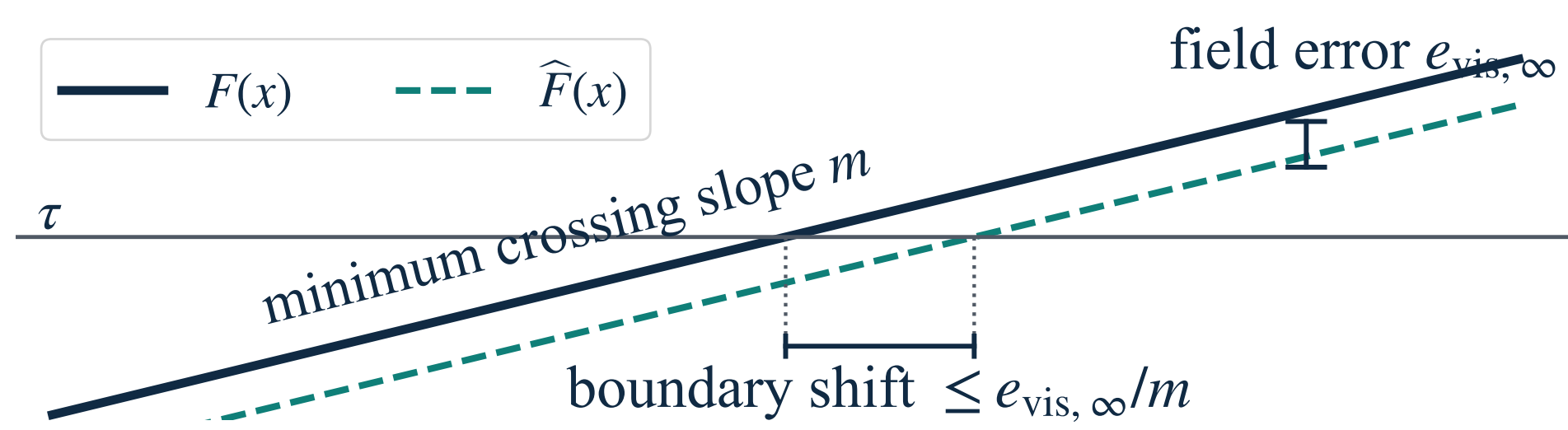


The displayed raw powers are near-sharp benchmark exponents; the general upper bound has an arbitrarily small exponent loss. Only the fixed-scale rank tail is exponentially small; noise still limits the full reconstruction.

04 Geometry stability

Set $F = H_\ell f^\dagger$ and $\hat{F} = H_\ell f_{\text{rec}}$; write $\Gamma = \Gamma_\tau(F)$ and $\hat{\Gamma} = \Gamma_\tau(\hat{F})$. For a compact C^1 interface and continuous $\hat{F}$, under the signed-transversality, valid-tube, and separation assumptions in [4],

$$e_{\text{vis},\infty} \leq \varepsilon_{\text{vis}} < ma \implies d_H(\hat{\Gamma}, \Gamma) \leq \frac{e_{\text{vis},\infty}}{m} \leq \frac{\varepsilon_{\text{vis}}}{m}.$$



A transverse crossing converts pointwise field error into controlled boundary displacement.

CORE MECHANISM



At a declared physical scale, parabolic inverse reconstruction can preserve geometry far more accurately than its raw pixelwise values.

RAW

Discontinuous source recovery is algebraically slow in L^2 .

VISIBLE

At fixed ℓ , the unresolved spectral rank tail is exponentially small.

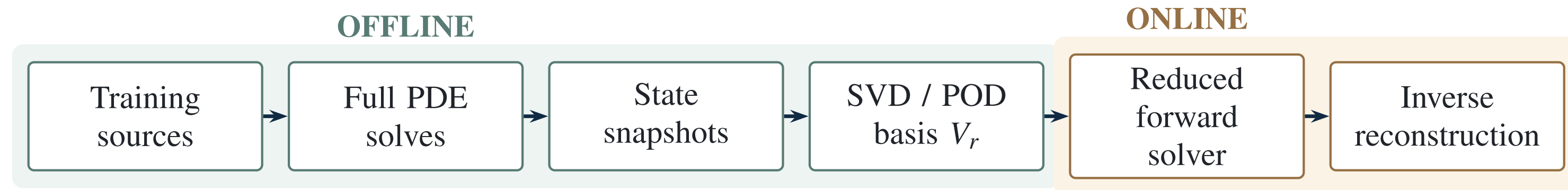
GEOMETRY

Visible L^∞ error plus a level-set margin yields Hausdorff control.

References

- [1] Wang, Zhang & Zhang. *J. Comput. Phys.* 487 (2023), 112156.
[2] Engl, Hanke & Neubauer. *Regularization of Inverse Problems*. Kluwer, 1996.

POD acceleration

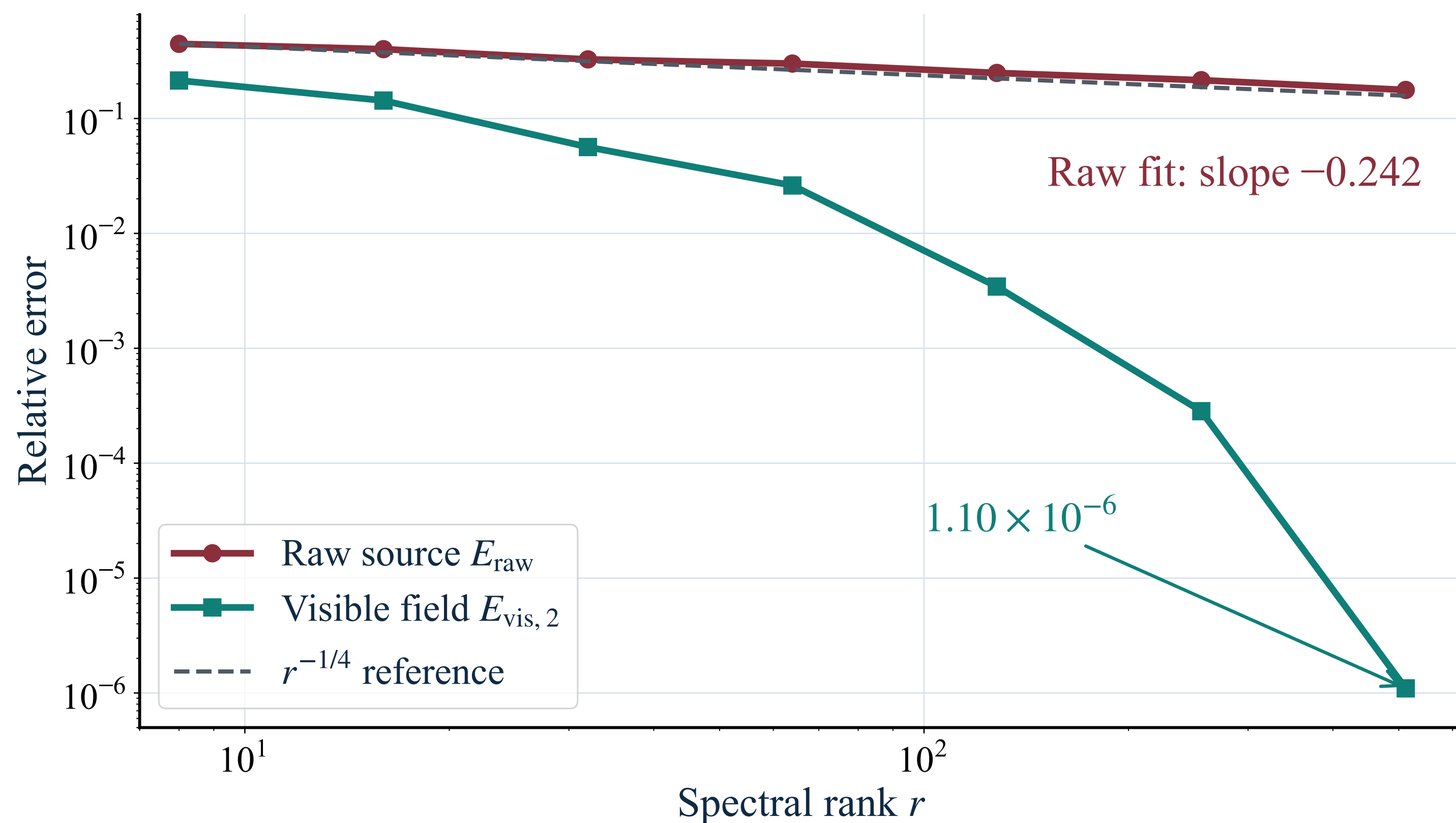


POD learns a low-dimensional state basis offline; the reduced solver then makes repeated forward evaluations cheaper [1,3].

Reduced reconstructions motivate the question: why can coarse shape remain recognizable despite substantial raw source error?

05 Numerical evidence

The raw error follows the predicted algebraic rank trend, while the fixed-scale visible error decreases by more than five orders of magnitude.



RAW FIT

$$-0.242 \approx -1/4$$

VISIBLE ERROR REDUCTION

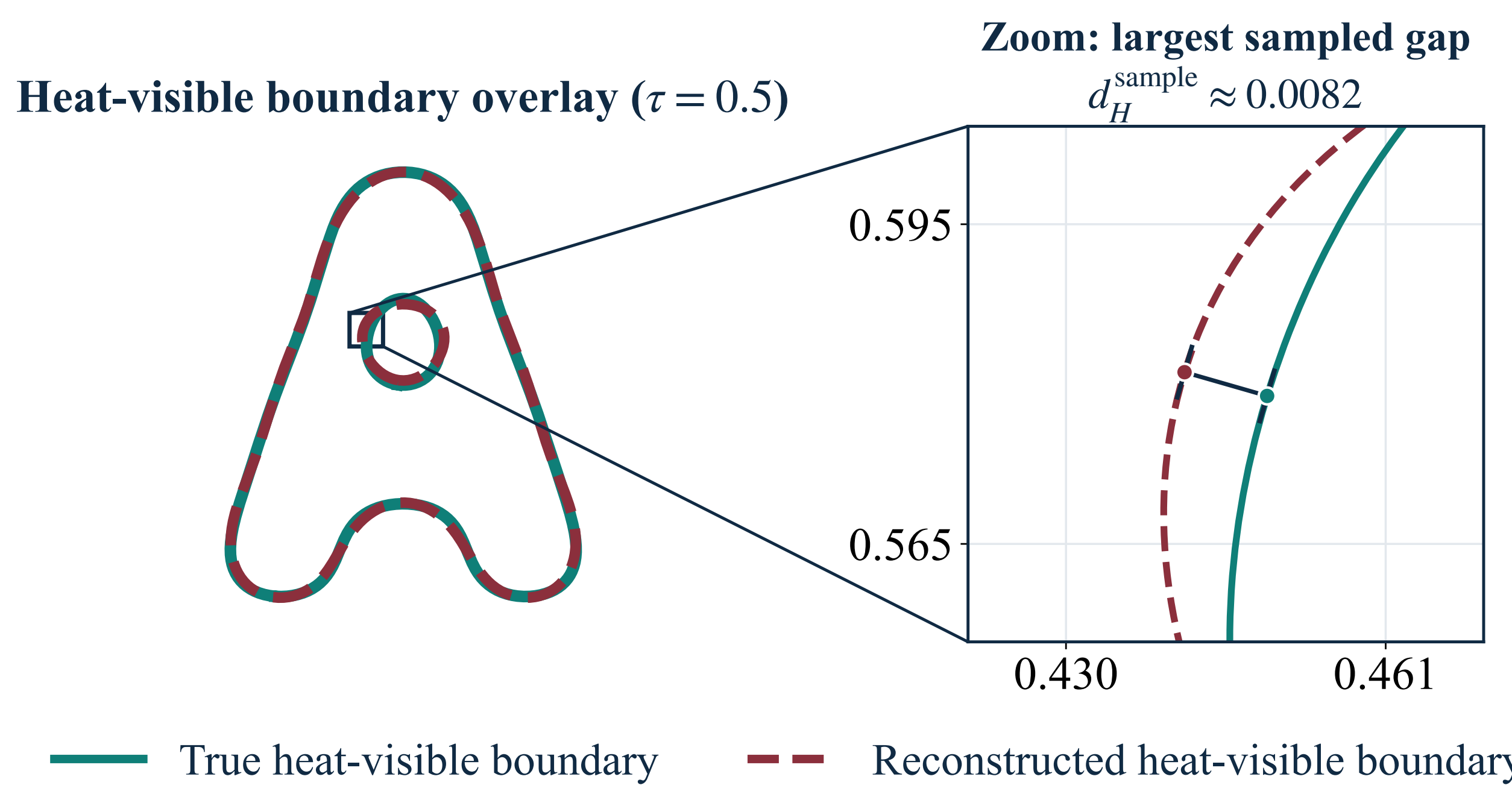
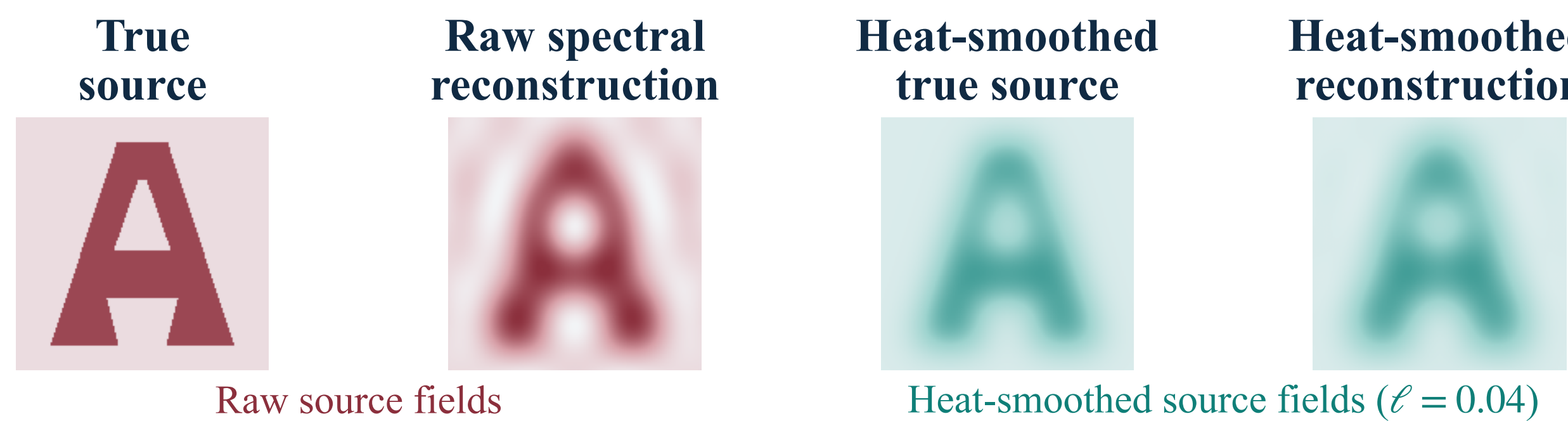
$$2.14 \times 10^{-1} \implies 1.10 \times 10^{-6}$$

> 5 orders of magnitude reduction

Fig. 1. Rectangular indicator on a 96^2 grid, $T = 0.15$, $\Delta t = 0.005$, exact data, $\lambda = 0$ spectral truncation, and $\ell = 0.04$. This plot tests rank separation, rather than the λ -rate theorem; fitted slopes are consistency diagnostics, not proofs.

06 Visible shape recovery

A larger-hole letter A tests nonconvex visible geometry. Here H_ℓ smooths the source, not the final-time state $S_T f$.



$$E_{\text{raw}} = 0.417 \quad E_{\text{vis},2} = 0.040 \quad e_{\text{vis},\infty} = 0.051$$
$$d_H^{\text{sample}} = 0.0082 < e_{\text{vis},\infty}/m_{\text{num}} = 0.0166, \text{ with } m_{\text{num}} = 3.09 \text{ (sampled consistency diagnostic).}$$

Fig. 2. Exact-data spectral projection $f_{\text{rec}} = P_r f^\dagger$ on a 192^2 grid ($r = 72$, $\ell = 0.04$, $\tau = 0.5$). The sampled relation $d_H^{\text{sample}} = 0.0082 < e_{\text{vis},\infty}/m_{\text{num}} = 0.0166$ is consistent with the level-set estimate, but does not certify all continuum tube assumptions. Since the source has corners, no global $O(\ell^2)$ claim is made.

- [3] Hu, Zhang & Zhu. *Inverse Problems* 41 (2025), 055015.

- [4] Chow. Raw, Heat-Visible, and Geometric Error in Parabolic Inverse Reconstruction of Indicator Sources. Working manuscript.